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Handgrip strength parameters as predictors of post-COVID syndrome severity and muscle weakness in an observational study

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Post-COVID Syndrome (PCS) presents with persistent symptoms such as fatigue, muscle weakness, and cognitive impairments, which often impact quality of life. This study determines handgrip strength (HGS) using a digital hand dynamometer as an objective measure of muscle function in PCS, aiming to assess its utility in symptom quantification and explore its relationship with endothelial dysfunction (ED) and inflammatory markers. In a cohort of 102 patients with PCS (77 women and 25 men), 62 of 101 classified patients (61.4%) fulfilled criteria for myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS). HGS parameters, including mean HGS, fatigability, and recovery, were assessed. Patients with ME/CFS exhibited lower mean HGS ($p = 0.031$), greater fatigability ($p = 0.042$), and poorer recovery ($p = 0.048$) compared with patients without ME/CFS. Furthermore, HGS parameters were significantly correlated with PCS severity and patient-reported outcome measures (including COVID-19 Yorkshire Rehabilitation Scale, PCS Severity Score and Fatigue Severity Scale). A gender-specific analysis revealed associations between HGS and ferritin in males ($r = 0.42$, $p = 0.036$) and thrombocyte levels inversely related to muscle recovery ($r = -0.43$, $p = 0.030$), potentially indicating microvascular compromise. Elevated intercellular adhesion molecule-1 (ICAM-1) levels were observed in patients with reduced recovery and VEGF was inversely correlated with fatigability in ME/CFS patients ($r = -0.26$, $p = 0.044$), suggesting that ED is involved in PCS-related muscle dysfunction. Anti-S1 IgG titers correlated negatively with HGS in females ($r = -0.24$, $p = 0.040$), suggesting a potential role of viral persistence in muscle impairment. These findings underscore the role of HGS in PCS assessment, revealing phenotype- and gender-specific impairments and highlighting the need for further research into ED pathways and targeted rehabilitation strategies to enhance PCS management.

Keywords Post-COVID syndrome, Handgrip strength, Fatigue, Long COVID, Endothelial dysfunction

In the aftermath of COVID-19, a considerable cohort of individuals continue to struggle with a complex array of symptoms, extending beyond the acute phase of the infection. This condition is recognized as Post-COVID Syndrome (PCS) or Long COVID[1,2]. PCS is defined by the World Health Organization (WHO) as symptoms persisting for a minimum of two months within three months following a SARS-CoV-2 infection; these symptoms must affect daily functioning and cannot be explained by an alternative diagnosis[3]. With an estimated incidence of 10–20% and more than 651 million documented SARS-CoV-2 cases worldwide, PCS presents immense challenges to healthcare systems worldwide[4–7].

PCS is characterized by symptoms such as fatigue, myalgia, exertional intolerance, postexertional malaise (PEM), cognitive and autonomic dysfunction, and headaches[1,8]. Within this spectrum, a large subset of patients

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meets the diagnostic criteria for postinfectious myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)[9,10]. The prevalence of postinfectious ME/CFS was substantial even before the COVID-19 pandemic, affecting approximately 3 million individuals on the European continent. With the increased incidence of COVID-19, the emergence of ME/CFS after infection poses a considerable challenge for healthcare and society[11].

Endothelial dysfunction (ED) and persistent inflammation are increasingly recognized as critical factors in the long-term complications of COVID-19 and ME/CFS[12,13]. Adhesion molecules such as ICAM-1 and VCAM-1 play central roles by mediating leukocyte adhesion and migration, contributing to endothelial inflammation in patients and leading to impaired lung function and vascular damage in PCS[14–18]. ED may lead to muscle weakness by reducing microvascular blood flow, thereby limiting oxygen and nutrient delivery to muscle tissues and causing muscle fatigue and decreased strength. This suggests that ED could be a contributing factor to persistent symptoms such as muscle weakness and fatigue in PCS[19]. Given the potential role of ED in PCS-related functional impairment, objective measures of muscle performance are of particular interest.

Handgrip strength (HGS) is an objective measure of muscle strength widely studied in sarcopenia as a predictor of mortality and morbidity[20–25]. Given the role of muscle weakness in PCS, HGS is increasingly relevant and is associated with poorer functional outcomes[26,27]. Accordingly, patients with PCS are expected to exhibit reduced handgrip strength, with impairments extending beyond normal age- and gender-related differences in muscle function. While sarcopenia denotes age-related muscle loss, the term dynapenia refers to reduced muscle strength and power and is preferred in the context of PCS[28].

While previous studies have examined the associations between functional outcomes and handgrip strength (HGS) in patients with PCS, an in-depth characterization of patient-reported outcomes (PROMs), which capture patients' subjective symptom burden, functional limitations, and quality of life through validated questionnaires, and their specific relationships with HGS is still needed.

Therefore, the primary aim of this study was to characterize handgrip strength (HGS) parameters, with a particular focus on muscle recovery, as objective measures of muscle function in patients with PCS and to relate these to clinical phenotypes, including ME/CFS. A comprehensive set of PROMs was collected, and two distinct questionnaires were employed to assess PCS severity. As a secondary aim, we investigated associations between HGS parameters and biomarkers of endothelial dysfunction and inflammation, to better understand potential mechanisms contributing to muscle weakness in PCS.

Materials and methods

Study design and cohort

This study is part of the "All Eyes on PCS" study, a prospective, observational single-center study focused on examining the retinal microvasculature of PCS patients and providing a comprehensive clinical characterization. A detailed description of patient recruitment and clinical characterization can be found in the study protocol published previously[29]. Patients were eligible if they had a confirmed SARS-CoV-2 infection (PCR or antibody test ≥ 3 months prior) and persistent PCS-typical symptoms lasting ≥ 2 months without an alternative explanation. Exclusion criteria comprised missing consent, age < 18 years, pregnancy, malignancy, conditions markedly reducing life expectancy, rheumatological autoimmune disease, and ocular or neurological disorders (cataract, glaucoma, epilepsy). Between October 2022 and September 2023, 105 patients were recruited via social media (76/105) and from the PCS outpatient department (29/105). Three patients were excluded from the data analysis (one due to a lack of proof of infection, one due to a lack of temporal association between SARS-CoV-2 infection and the onset of PCS symptoms, and one because they no longer exhibited PCS-typical symptoms on the day of recruitment). Data were collected at TU Munich (Departments of Nephrology, Neurology, and Ophthalmology) by experienced examiners who were aware of the study context but not involved in hypothesis formulation or statistical analyses.

The study protocol received approval from the local ethics committee (Ethics Committee of the Technical University of Munich, School of Medicine, Klinikum rechts der Isar; Approval number: 2022-317-S-SR) and was registered in advance (https://clinicaltrials.gov/ct2/show/NCT05635552). All procedures were conducted in accordance with the Declaration of Helsinki and relevant institutional guidelines and regulations. Written informed consent was obtained from all participants.

The final analysis cohort comprised 102 patients with PCS. Median age was 41.5 years (IQR 32.0–52.0), and 77 participants (75.5%) were female and 25 (24.5%) male. Median symptom duration at the time of assessment was 404 days (IQR 291.0–566.0). Most patients had experienced a mild to moderate acute infection not requiring hospitalization (59.8% severity grade 2), and 62 of 101 classified patients (61.4%) fulfilled the diagnostic criteria for ME/CFS. Baseline characteristics of the full cohort are provided in supplementary Table 1.

Sample size

The present work is a secondary analysis of the prospective "All Eyes on PCS" study. The sample size of this study has been calculated for the primary outcome concerning retinal vessel analysis[29]. This study was designed as a case–control study comparing patients with PCS to recovered individuals without persistent symptoms, and a target of 100 participants per group was derived from published normative data and expected between-group differences in retinal vessel parameters, allowing for potential dropouts. The present analysis includes the PCS group only (n = 102). No a priori sample size calculation was performed for HGS.

Handgrip strength measurement

The handgrip strength (HGS) of the dominant hand was assessed via a digital hand dynamometer (Jamar) across two separate sessions. Each session was separated by a 60-min rest period during which participants engaged in no strenuous physical activity. The Jamar dynamometer is widely regarded as the gold standard for HGS assessment, with excellent validity and high test–retest and inter-rater reliability in both clinical and research

settings[20]. To minimize learning effects and motivational bias, participants were familiarized with the device and measurement procedure prior to testing. During each session, participants were seated upright with elbows at 90° flexion and forearms fully supinated. All measurements were conducted under standardized conditions by trained examiners, who provided identical verbal instructions and standardized verbal encouragement to ensure consistent maximal effort. Each contraction lasted three seconds, followed by a five-second rest, repeated ten times per session. The device automatically recorded the peak value of each contraction (kg). To estimate strength, fatigability, and recovery, we analyzed various parameters and ratios as previously described[30], as outlined in Table 1. Normative values for Fmax were adjusted for age and gender[31].

Patient-related outcome measures

COVID-19 Yorkshire Rehabilitation Scale (C19-YRS)

The C19-YRS has 22 items, each rated on an 11-point scale from 0 (no symptoms) to 10 (extremely severe). It is divided into four subscales: symptom severity (0–100), functional disability (0–50), additional symptoms (0–60), and overall health (0–10)[32].

PCS Severity Score

Twelve typical PCS symptoms are assessed and multiplied by a predetermined coefficient. This results in a total score ranging from 0 to 59 points. The higher the score is, the greater the PCS severity. The classification is divided into three groups: none/mild PCS (≤ 10.75), moderate PCS (> 10.75 and ≤ 26.25), and severe PCS (> 26.25)[33].

Fatigue Severity Scale (FSS)

This questionnaire covers nine signs and symptoms of fatigue, scored from 1 to 7. Therefore, a score of 9–64 points can be given; the higher the score is, the more severe the symptoms are[34].

9-item Patient Health Questionnaire (PHQ-9)

Recommended by many international guidelines, the PHQ-9 is a very reliable screening tool for depression. Nine symptoms of depression within the past two weeks were queried[35].

EuroQol 5-Dimensions 5-Levels questionnaire (EQ-5D-5L)

Health-related quality of life was assessed via the EQ-5D-5L questionnaire, which evaluates five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression, each rated on a five-level scale[36].

Laboratory values

Standard laboratory measurements were conducted in an ISO-certified laboratory. The serum levels of IL-6, IL-8, CXCL10, Rantes, ICAM, VCAM, VEGF, and MCP-1 were determined following previously described methods[13]. For the detection of von Willebrand factor (vWF) antigen (Ag) in human samples, we utilized the LIAPHEN vWF:Ag kit (HYPHEN BioMed, Neuville-sur-Oise, France), which is an immunoturbidimetric assay. To measure the levels of SARS-CoV-2 spike IgG in plasma samples, we employed an enzyme-linked immunosorbent assay (ELISA). Specifically, the LEGEND MAX ELISA kit from BioLegend was utilized, following the protocol provided by the manufacturer.

Definition of subgroups

Four subgrouping variables were used to define exploratory, hypothesis-generating subgroup analyses:

Recovery ratio

The recovery ratio (Fmean2/Fmean1, see Table 1) reflects the restoration of muscle strength after a 60-min rest and thus corresponds most closely to the delayed recovery after exertion that characterizes PEM[30]. As a within-patient ratio it is largely independent of age, gender, body composition and absolute strength, and it has an intrinsic physiological reference point close to 1.0 in healthy individuals[30], which makes it suitable for dichotomization. As no established threshold for the recovery ratio exists, patients were dichotomized into abnormal (≤ 0.8, n = 23) and normal (> 0.8, n = 79) recovery; this data-driven cutoff approximately corresponds to the lower quartile of our cohort, and sensitivity analyses using 0.75 and 0.85 are reported in supplementary Table 8.

Parameter	Formula	Explanation
Fmax	Fmax in kg adjusted for gender and age	Maximum strength of both sessions
Fmean	$\frac{\sum 10\ \text{trials}}{10}$	Mean grip strength of all 10 trials
Fatigue ratio	$\frac{\text{Fmax}}{\text{Fmean}}$	Higher values indicate fatigability
Recovery ratio	$\frac{\text{Fmean2}}{\text{Fmean1}}$	Low values indicate impaired recovery

Table 1. Parameters of muscle strength. Fmax = maximum handgrip strength, Fmean = mean handgrip strength.

ME/CFS
ME/CFS was diagnosed according to the Canadian Consensus Criteria[37] and patients were grouped by its presence (n = 62) or absence (n = 39). In one patient, the diagnostic criteria were not correctly applied, and the patient was therefore excluded from this classification.

Gender
Given known gender differences in muscle strength and the female predominance of PCS, correlations with laboratory markers were analyzed separately in male (n = 25) and female (n = 77) patients and tested for effect modification using age-adjusted interaction models.

Depression
Patients were stratified post hoc by PHQ-9 depression severity into 4 groups (PHQ-9 < 10: no depression, 10–14: mild depression, 15–19: moderate depression, ≥ 20: severe depression) to address whether reduced HGS reflects lack of drive rather than physiological impairment.

Statistical analysis
The statistical analysis was performed using R (version 4.2.1) and RStudio (version 2024.04.2 + 764) using the packages *dplyr*, *tidyr*, *purrr*, *broom*, *pROC*, *ggplot2*, *car*, and *gt*. Normally distributed values are presented as the means ± SDs, whereas nonnormally distributed values are presented as medians and interquartile ranges (IQRs). Categorical variables are presented as counts and percentages. Normality was analyzed via graphical analysis with histogram plotting and boxplotting and with the Shapiro–Wilk test for normality. For comparisons of baseline characteristics, the Welch two-sample test was used for normally distributed variables, the Wilcoxon rank sum test was used for nonnormally distributed variables, and the chi-square test was used for categorical values. The ggpubr package was used to calculate Spearman´s and Pearson´s correlation coefficients to generate scatterplots. All models were specified a priori based on clinical relevance and previous literature. Age and gender were included as covariates in all multivariable models to account for known effects on muscle strength. No automated stepwise selection procedures were applied. For all logistic and linear regression models, standard diagnostic procedures were performed, including inspection of residuals, assessment of influential points (Cook's distance), and evaluation of model fit (Hosmer–Lemeshow test for logistic models). No major violations of model assumptions were detected. Diagnostic plots for a representative model are shown in supplementary Fig. 1; similar results were observed for the remaining models. To examine the potential discriminatory ability of HGS parameters, receiver operating characteristic analysis was performed via the plotROC and pROC packages. The correlation matrix was created via the corrplot package. These analyses were exploratory in nature and intended to generate hypotheses for future studies rather than provide confirmatory evidence. We used the interaction package to generate the interaction plots; the $p_{interact}$ was calculated via a multivariate linear regression model. All graphical plots were created using the ggplot2 package. All clinical data were typed by two independent researchers, who checked for mistakes.

Results
Patient characteristics
A total of 102 patients with PCS with a previous history of SARS-CoV-2 infection and ongoing symptoms were recruited (see baseline characteristics of the full cohort in supplementary Table 1). Patients were categorized based on an abnormal recovery ratio, with a cutoff value of 0.8. Among them, 23 patients (mean age 44.1 ± 10.2 years, 82.6% female) had an abnormal recovery ratio, whereas 79 patients (mean age 41.2 ± 11.9 years, 73.4% female) had a normal recovery ratio. Concerning cardiovascular risk factors, PCS patients with abnormal recovery had significantly higher cholesterol levels ($p = 0.03$). In addition, we observed higher levels of ICAM-1 in PCS patients with abnormal recovery ($p = 0.014$). There was a tendency toward a history of more severe acute SARS-CoV-2 infection; however, this difference was not significant ($p = 0.08$). Interestingly, there was no significant difference in the duration of PCS. Further detailed demographic and clinical characteristics of the cohort are presented in Table 2, laboratory values of the cohort are shown in Table 3.

HGS is correlated with PCS severity, C19-YRS score, and PROMs
Figure 1 shows the different measures of HGS and their Pearson correlations with PCS severity scores and patient-reported outcome measures (PROMs). Patients with lower mean handgrip strength (Fmean) values presented more severe symptoms, as measured by the PCS severity score (r = − 0.31, 95% CI − 0.48 to − 0.12, $p = 0.001$) and the COVID-19 Yorkshire rehabilitation scale (C19-YRS; r = − 0.30, 95% CI − 0.47 to − 0.11, $p = 0.004$). Additionally, the fatigability (fatigue ratio) was significantly correlated with the PCS severity score (r = 0.22, 95% CI 0.03 to 0.40, $p = 0.028$) and with the C19-YRS (r = 0.25, 95% CI 0.06 to 0.42, $p = 0.019$). Higher PCS severity scores (r = − 0.15, 95% CI − 0.33 to 0.05, $p = 0.121$) and higher C19-YRS scores (r = -0.17, 95% CI − 0.35 to 0.03, $p = 0.104$) were negatively correlated with recovery, but these correlations were not statistically significant. The C19-YRS and PCS severity scores were positively correlated (r = 0.40, 95% CI 0.21 to 0.57, $p < 0.001$). For the remaining PROMs (EQ-5D-5L, PHQ-9, FSS, and GAD-7), significant correlations were found for Fmean with FSS (r = -0.36, 95% CI − 0.52 to − 0.17, $p < 0.001$) and PHQ-9 (r = − 0.30, 95% CI − 0.46 to − 0.10, $p = 0.004$) as well as for the fatigue ratio with the FSS (r = 0.28, 95% CI 0.09 to 0.46, $p = 0.004$), PHQ-9 (r = 0.28, 95% CI 0.09 to 0.46, $p = 0.004$), and EQ-5D-5L (r = − 0.33, 95% CI − 0.58 to − 0.06, p = 0.025, see supplementary Fig. 2). To control for potential confounders of HGS, we fitted a multivariate linear model. The associations between higher severity, measured by the PCS severity score and C19-YRS, with Fmean and fatigue ratio, remained significant after adjustments were made (supplementary Table 2). These findings demonstrate significant associations between objective muscle strength measurements and subjective symptom severity measured by PROMs in PCS patients.

Demographic and clinical characteristics			
	Abnormal recovery (n = 23)	**Normal recovery (n = 79)**	***P* value**
Age			
Mean, SD	44.1 (± 10.2)	41.2 (± 11.9)	0.30
Gender			
Female	19 (82.6%)	58 (73.4%)	0.42
BMI			
kg/m^2, Mean, SD	24.3 (± 4.8)	24.6 (± 4.8)	0.79
Cardiovascular risk factors (CVRF)			
Nicotine abuse	4 (17.4%)	11 (13.9%)	0.74
Arterial hypertension	6 (26.1%)	10 (12.7%)	0.19
Hypercholesterolemia	10 (43.5%)	14 (17.7%)	**0.023***
DM	1 (4.3%)	0 (0.0%)	0.23
Acute infection and PCS characteristics			
Number of infections			0.40
0	0 (0.0%)	1 (1.3%)	
1	16 (69.6%)	54 (68.4%)	
2	6 (26.1%)	23 (29.1%)	
3	0 (0.0%)	1 (1.3%)	
4	1 (4.3%)	0 (0.0%)	
Severity acute infection			0.08
Asymptomatic	1 (4.3%)	0 (0.0%)	
Symptomatic, independent	11 (47.8%)	50 (63.3%)	
Symptomatic, assisted	10 (43.5%)	24 (30.4%)	
Hospitalization no oxyg.	0 (0.0%)	4 (5.1%)	
Hospitalization oxyg. mask	1 (4.3%)	0 (0.0%)	
Hospitalization high flow	0 (0.0%)	1 (1.3%)	
Number of vaccinations			0.39
0	0 (0.0%)	5 (6.3%)	
1	1 (4.3%)	3 (3.8%)	
2	9 (39.1%)	18 (22.8%)	
3	9 (39.1%)	44 (55.7%)	
4	3 (13.0%)	8 (10.1%)	
5	1 (4.3%)	1 (1.3%)	
SARS-CoV-2 Variant			0.40
Alpha	3 (13.0%)	4 (5.1%)	
Delta	1 (4.3%)	12 (15.2%)	
Omikron	5 (21.7%)	14 (17.7%)	
Unknown	14 (60.9%)	49 (62.0%)	
PCS duration (months)			
Median, IQR	11.0 (7.0—20.0)	13.0 (9.0—17.5)	0.72
Sick days			
Mean, SD	192.0 (± 148.7)	270.3 (± 255.1)	0.16
Preexisting illness			
Depression	5 (21.7%)	15 (19.0%)	0.83
Bronchial asthma	4 (17.4%)	15 (19.0%)	1.0
Hypothyroidism	3 (13.0%)	17 (21.5%)	0.53

Table 2. Demographic and clinical characteristics of PCS patients with abnormal (≤ 0.8, n = 23) versus normal (> 0.8, n = 79) recovery. Continuous variables are given as mean (± SD) or median (IQR), categorical variables as n (%). *P* values: t-test (parametric), Wilcoxon rank sum test (nonparametric), χ^2 test (categorical), Fisher's exact test (proportional); for categorical variables a single p value applies to the variable as a whole. Hypercholesterolemia is defined as cholesterol > 200 mg/dl. Laboratory parameters are shown in Table 3. *$p < 0.05$. BMI = body mass index; CVRF = cardiovascular risk factors; DM = diabetes mellitus type I or II; IQR = interquartile range; PCS = post-COVID syndrome; SD = standard deviation. Significant values are in [bold].

Laboratory parameters			
Leukocytes			
G/L Mean (SD)	6.9 (± 2.1)	6.6 (± 1.8)	0.53
Neutrophils			
%, Mean (SD)	62.2 (± 7.8)	62.6 (± 8.6)	0.85
Lymphocytes			
%, Mean (SD)	28.5 (± 6.6)	27.9 (± 8.0)	0.72
Monocytes			
%, Median (IQR)	7.0 (6.0–8.0)	7.0 (6.0–8.5)	0.79
Ferritin			
ng/ml, Median (IQR)	91.0 (65.0–140.0)	76.0 (44.0–157.2)	0.71
CRP			
mg/dl, Median (IQR)	0.1 (0.1–0.1)	0.1 (0.1–0.2)	0.79
IL-6			
pg/ml, Median (IQR)	2.2 (1.2–4.1)	2.1 (0.8–4.2)	0.92
CXCL10			
pg/ml,Median (IQR)	57.1 (39.9–69.8)	54.5 (42.2–65.4)	0.99
Rantes			
pg/ml, Mean (SD)	1922.4 (± 1054.8)	1794.9 (± 837.7)	0.55
MCP-1			
pg/ml, Median (IQR)	18.0 (15.0–20.6)	15.3 (10.6–21.8)	0.24
IL-8			
fg/ml, Median (IQR)	1060.9 (866.1–1417.5)	1135.2 (768.6–1848.5)	0.94
VEGF			
pg/ml, Median (IQR)	2.6 (0.4–9.4)	4.5 (0.4–8.3)	0.88
Thrombocytes			
G/L, Mean (SD)	308.0 (± 53.2)	282.4 (± 64.8)	0.061
D-Dimer			
(µg/I), Median (IQR)	262.0 (199.0–366.0)	226.0 (199.0–393.0)	0.75
ICAM-1			
pg/ml, Median (IQR)	5751.7 (3928.8–15,296.8)	3958.8 (2143.8–5961.1)	**0.014***
VCAM-1			
pg/ml, Median (IQR)	7089.2 (0.0–13,505.2)	4761.5 (0.0–11,479.1)	0.50
vWF			
%, Median (IQR)	124.6 (85.1–173.0)	108.4 (82.8–156.5)	0.49
aPTT			
Sec, Mean (SD)	29.3 (± 3.4)	30.0 (± 3.0)	0.39
hs Troponin T			
ng/l, Median (IQR)	0.0 (0.0–0.0)	0.0 (0.0–0.0)	0.35
Hemoglobin			
g/dl, Mean (SD)	13.6 (± 1.0)	13.8 (± 1.3)	0.55
Transferrin			
mg/dl, Mean (SD)	271.1 (± 26.3)	257.8 (± 34.6)	0.093
Saturation of transferrin			
%, Mean (SD)	25.3 (± 10.6)	28.6 (± 10.9)	0.21
Red cell distribution width			
%, Median (IQR)	12.5 (12.1–13.0)	12.6 (12.1–13.1)	0.79
MCHC			
g/dl, Mean (SD)	34.0 (± 0.9)	34.1 (± 1.1)	0.64
IgG1			
mg/dl, Mean (SD)	633.8 (± 140.6)	680.3 (± 135.4)	0.16
IgG2			
mg/dl, Mean (SD)	355.6 (± 95.5)	357.9 (± 119.4)	0.93
IgG3			
mg/dl, Median (IQR)	31.9 (19.9–41.0)	26.4 (19.4–37.2)	0.49
IgG4			
mg/dl, Median (IQR)	46.1 (28.2–61.0)	47.7 (25.8–82.6)	0.32
Continued			

Laboratory parameters			
IgG			
mg/dl, Mean (SD)	1036.0 (± 181.5)	1094.7 (± 210.0)	0.23
Cholesterol			
mg/dl,Mean (SD)	220.3 (± 53.3)	197.7 (± 40.1)	**0.03***
TSH			
µU/mL, Median (IQR)	1.6 (1.2–2.1)	1.4 (0.9–1.9)	0.23

Table 3. Laboratory parameters of PCS patients with abnormal (≤ 0.8, n = 23) versus normal (> 0.8, n = 79) recovery. Values are given as mean (± SD) or median (IQR). *P* values: t-test (parametric) and Wilcoxon rank sum test (nonparametric). *$p < 0.05$. aPTT = activated partial thromboplastin time; CRP = C-reactive protein; CXCL10 = C-X-C motif chemokine ligand 10; hs troponin T = high-sensitivity troponin T; ICAM-1 = intercellular adhesion molecule 1; IgG = immunoglobulin G; IL = interleukin; IQR = interquartile range; MCHC = mean corpuscular hemoglobin concentration; MCP-1 = monocyte chemoattractant protein 1; PCS = post-COVID syndrome; RANTES = regulated on activation, normal T cell expressed and secreted; SD = standard deviation; TSH = thyroid-stimulating hormone; VCAM-1 = vascular cell adhesion molecule 1; VEGF = vascular endothelial growth factor; vWF = von Willebrand factor. Significant values are in bold.

Handgrip strength is reduced in PCS patients with ME/CFS

Figure 2 shows the results of the various HGS measures based on ME/CFS status, as defined by the Canadian Consensus Criteria. Compared with those without ME/CFS (n = 39), PCS patients with ME/CFS (n = 62) presented significantly lower mean handgrip strength ($p = 0.031$), greater fatigability ($p = 0.042$), and poorer recovery ($p = 0.048$). The associations between ME/CFS and lower Fmean remained significant after controlling for age and gender. However, the associations between ME/CFS and the dependent variables, fatigue ratio and recovery, were weaker after adjustments were made (supplementary Table 3). These findings highlight greater impairment in muscle strength, fatigability, and recovery in the ME/CFS group.

Gender-specific correlations between inflammatory parameters, parameters of endothelial dysfunction, humoral antibodies, and HGS values

To further investigate reduced handgrip strength in PCS patients, HGS parameters were correlated with markers of inflammation and ED. For visualization, we used a correlation matrix to demonstrate Spearman correlation coefficients, as shown in Fig. 3. Two subgroups were formed to address potential gender differences: male and female PCS patients. These analyses are exploratory and intended to generate hypotheses for future studies rather than provide confirmatory evidence.

In male PCS patients, a positive correlation was observed between Fmean and ferritin (r = 0.42, 95% CI 0.01 to 0.71, $p = 0.036$; Fig. 3a). Additionally, among the ED markers, there was a negative correlation between thrombocytes and recovery (r = − 0.43, 95% CI − 0.72 to − 0.03, $p = 0.030$; Fig. 3b), suggesting a potential link between platelet counts and muscle recovery. In contrast, female PCS patients had no significant correlations with inflammatory markers (Fig. 3d) or ED markers (Fig. 3e). However, anti-S1-IgG titers were negatively correlated with Fmean (r = − 0.24, 95% CI − 0.44 to − 0.01, $p = 0.040$; Fig. 3f), indicating a potential role of viral persistence in muscle strength.

To validate gender-specific differences in the relationships between HGS and thrombocyte, ferritin, and anti-S1-IgG titers, we fitted an age-controlled regression model with interaction effects. Only for the positive correlation between Fmean and ferritin we observed a significant interaction between gender and ferritin ($\beta_{interact} = 0.06$, 95% CI 0.00 to 0.11, $p_{interact} = 0.041$). The negative correlation between recovery and thrombocytes was not significant ($\beta_{interact} = -0.001$, 95% CI − 0.003 to 0.001, $p_{interact} = 0.41$). Similarly, we found no gender-specific effect for the negative correlation between Fmean and anti-S1-IgG titers ($\beta_{interact} = -1.85 \times 10^{-5}$, 95% CI -1.69×10^{-3} to 1.65×10^{-3}, $p_{interact} = 0.99$; supplementary Fig. 3).

Across both subgroups, HGS measurements (Fmean, fatigue ratio, and recovery) were highly significantly correlated. These findings suggest that patients with lower Fmean values exhibit greater fatigability and poorer recovery, consistent with the expected associations with muscle function.

To further investigate the associations between HGS and laboratory variables, we fitted multivariate linear models to account for potential confounders. The associations between ferritin, anti-S1 IgG, and Fmean remained significant after controlling for age and gender. However, the association between recovery and thrombocytes became weaker after adjustment (supplementary Table 4).

Inflammatory parameters, parameters of endothelial dysfunction, humoral antibodies, and HGS values in PCS patients with ME/CFS

As we observed a more significant impairment of muscle strength in patients with ME/CFS, we conducted a correlation analysis between blood markers and HGS parameters in this subgroup (Fig. 4). No significant correlations were found between inflammatory markers and HGS parameters (Fig. 4a). However, we identified a negative correlation between VEGF levels and the fatigue ratio (r = − 0.26, 95% CI − 0.48 to − 0.01, $p = 0.044$; Fig. 4b). Additionally, the previously observed negative correlation between Fmean and Anti-S1 IgG levels was confirmed in this subgroup (r = − 0.28, 95% CI − 0.50 to − 0.03, $p = 0.027$; Fig. 4c). Again, we fitted a multivariate linear regression model to control for age and gender. After adjustment, the negative association

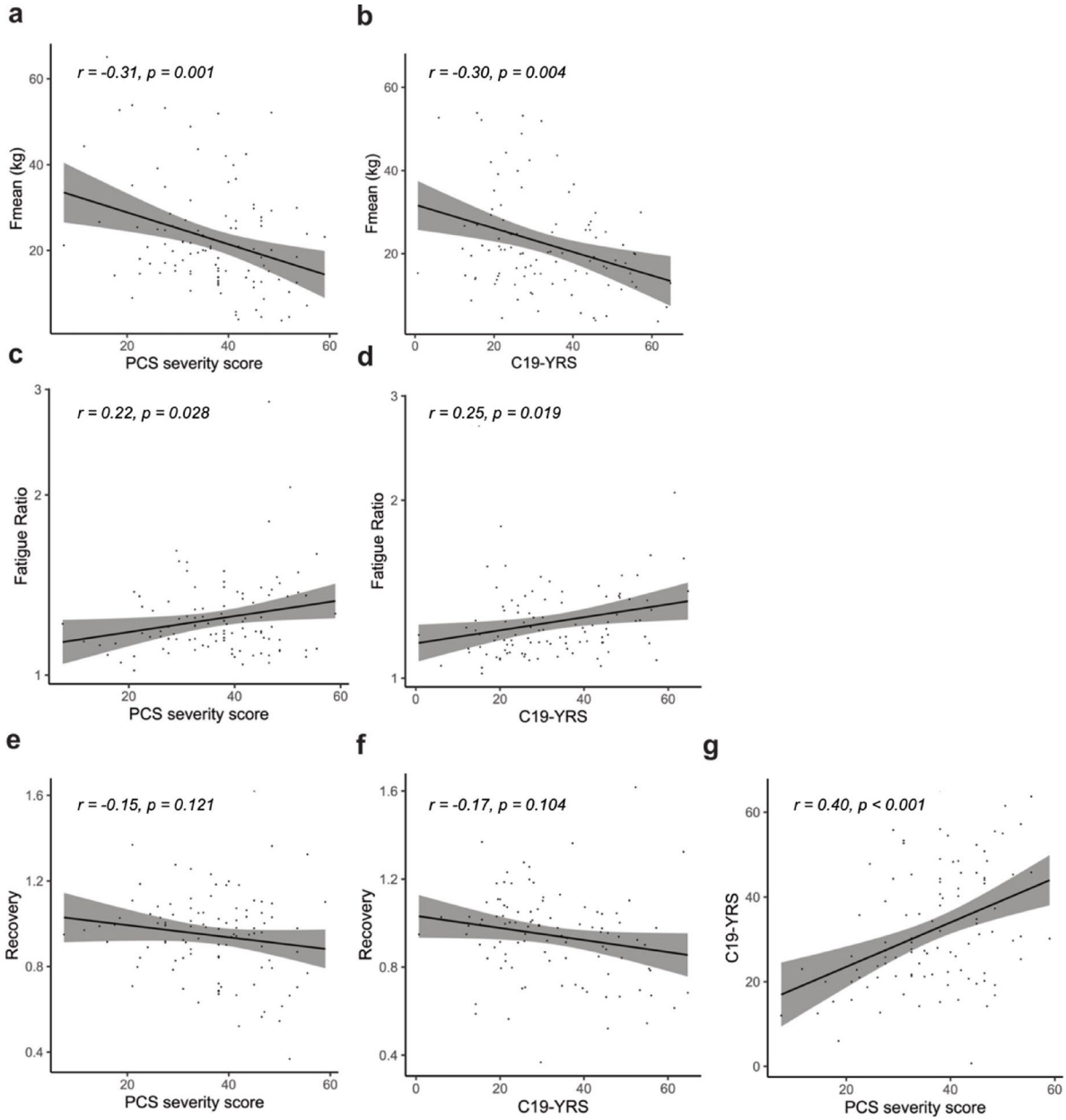


Fig. 1. Scatterplots show Pearson correlations between PCS severity scores (PCS severity score, n = 102; C19-YRS, n = 101) and HGS parameters (Fmean (kg), Fatigue ratio (Fmax/Fmean) and recovery (%, Fmean2/Fmean1)). Fatigue ratio is log-transformed. Plots show correlations with confidence intervals and the corresponding r and *p* value. Abbreviations: Fmean = mean handgrip strength, C19-YRS = COVID19-Yorkshire Rehabilitation Scale.

between anti-S1 IgG and Fmean was attenuated and no longer significant ($b = -8.0 \times 10^{-4}$, 95% CI -1.8×10^{-3} to 1.9×10^{-4}, $p = 0.11$). In contrast, an association between higher anti-S1 IgG and poorer recovery emerged after adjustment ($b = -2.3 \times 10^{-5}$, 95% CI -4.5×10^{-5} to -6.8×10^{-7}, $p = 0.043$). The observed correlation between the fatigue ratio and lower VEGF narrowly missed significance after controlling for confounders ($b = -5.7 \times 10^{-3}$, 95% CI -1.14×10^{-2} to 4.8×10^{-5}, $p = 0.052$), whereas higher VEGF levels were significantly associated with better recovery ($b = 5.4 \times 10^{-3}$, 95% CI 7.6×10^{-4} to 1.01×10^{-2}, $p = 0.023$). Surprisingly, after adjustment, we observed an association between higher levels of CXCL10 and Fmean (b = 0.07, 95% CI 0.005 to 0.131, $p = 0.036$; supplementary Table 5).

Discussion

The first aim of our study was to test whether HGS parameters can be used as an objective method to quantify symptom burden in PCS and to differentiate between PCS patients with and without ME/CFS. The second aim was to investigate the underlying pathomechanism of reduced muscle strength in patients with PCS. The

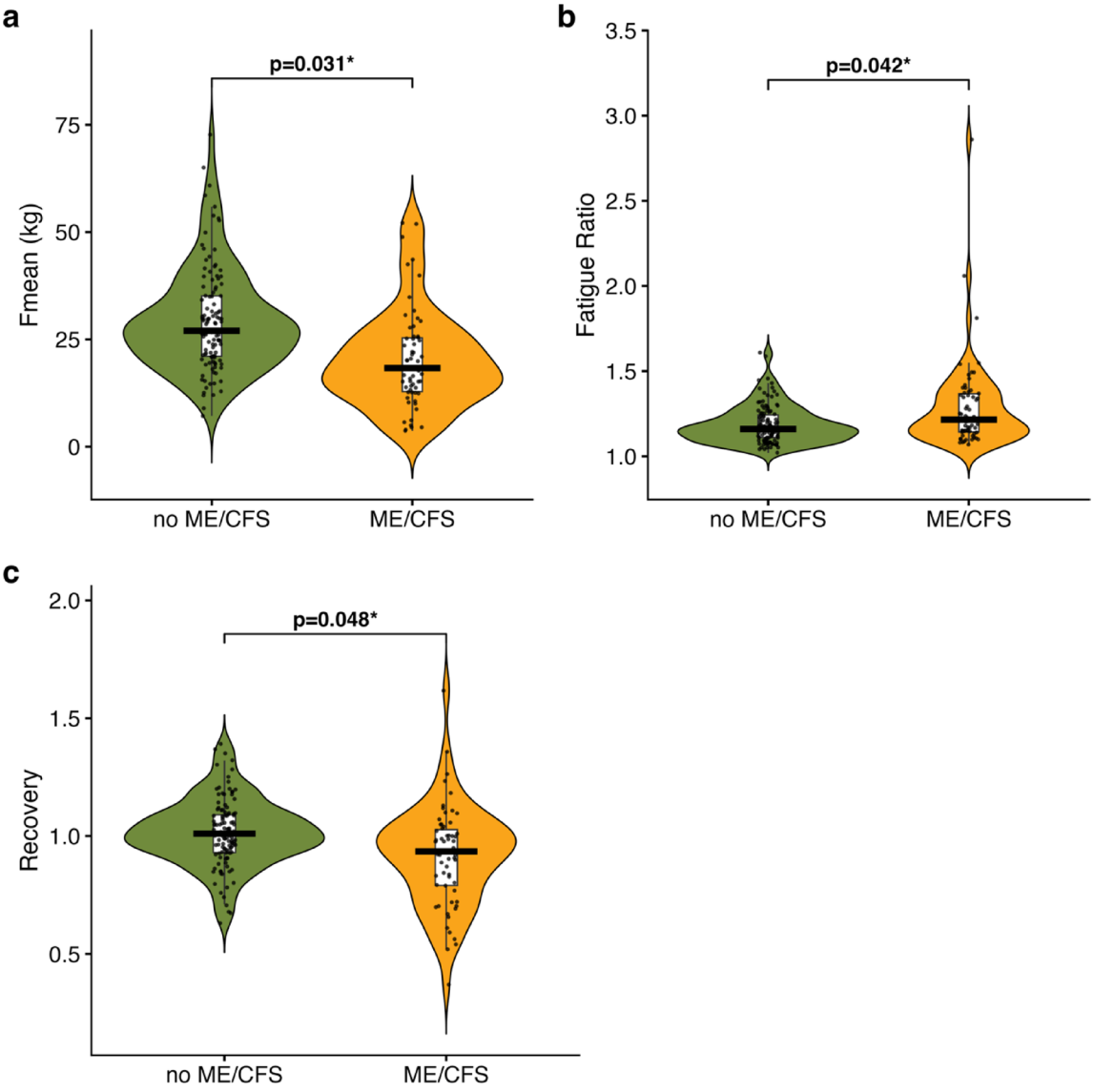


Fig. 2. Violin plots illustrate the distribution of Fmean (kg), Fatigue Ratio, and Recovery in individuals characterized by the presence (orange, n = 62) or absence (green, n = 39) of ME/CFS, groups are indicated on the x-axis. The central line represents the median and the box the interquartile range. Individual data points are shown as black dots scattered within the violins. P values were calculated using the Welch Two Sample t-test for parametric values. Abbreviations: Fmean = mean handgrip strength, ME/CFS = myalgic encephalomyelitis/chronic fatigue syndrome.

HGS parameters, particularly Fmean and fatigue ratio, were significantly correlated with both the PCS severity and C19-YRS scores, indicating that the HGS is a good measure of subjective complaints in all PCS patients. Additional PROMs, such as higher fatigability and lower quality of life, were significantly correlated with HGS. These correlations persisted even after adjusting for confounding variables, suggesting that HGS may serve as a reliable indicator of muscular impairments[38]. Compared with published reference data[31], our cohort showed reduced Fmean values, further supporting the presence of impaired muscle function in PCS patients (supplementary Table 6). However, it should be noted that the observed correlations were of small magnitude ($r \approx 0.2–0.3$), indicating that the predictive value of HGS for symptom severity is limited and should be interpreted with caution.

In a study conducted by Charité, impaired HGS in PCS was shown to be correlated with the severity of PCS-specific symptoms, but only in PCS patients with ME/CFS[39]. In our cohort, HGS was correlated with increased PCS severity in PCS patients, mostly independent of their ME/CFS status (supplementary Fig. 4). Possible explanations may be that the study from Berlin included only female patients and examined other, more ME/CFS-specific PROMs. Additionally, their observed correlations were not adjusted for potential confounding factors, such as age differences. Our findings align with a recent study demonstrating that lower HGS values are

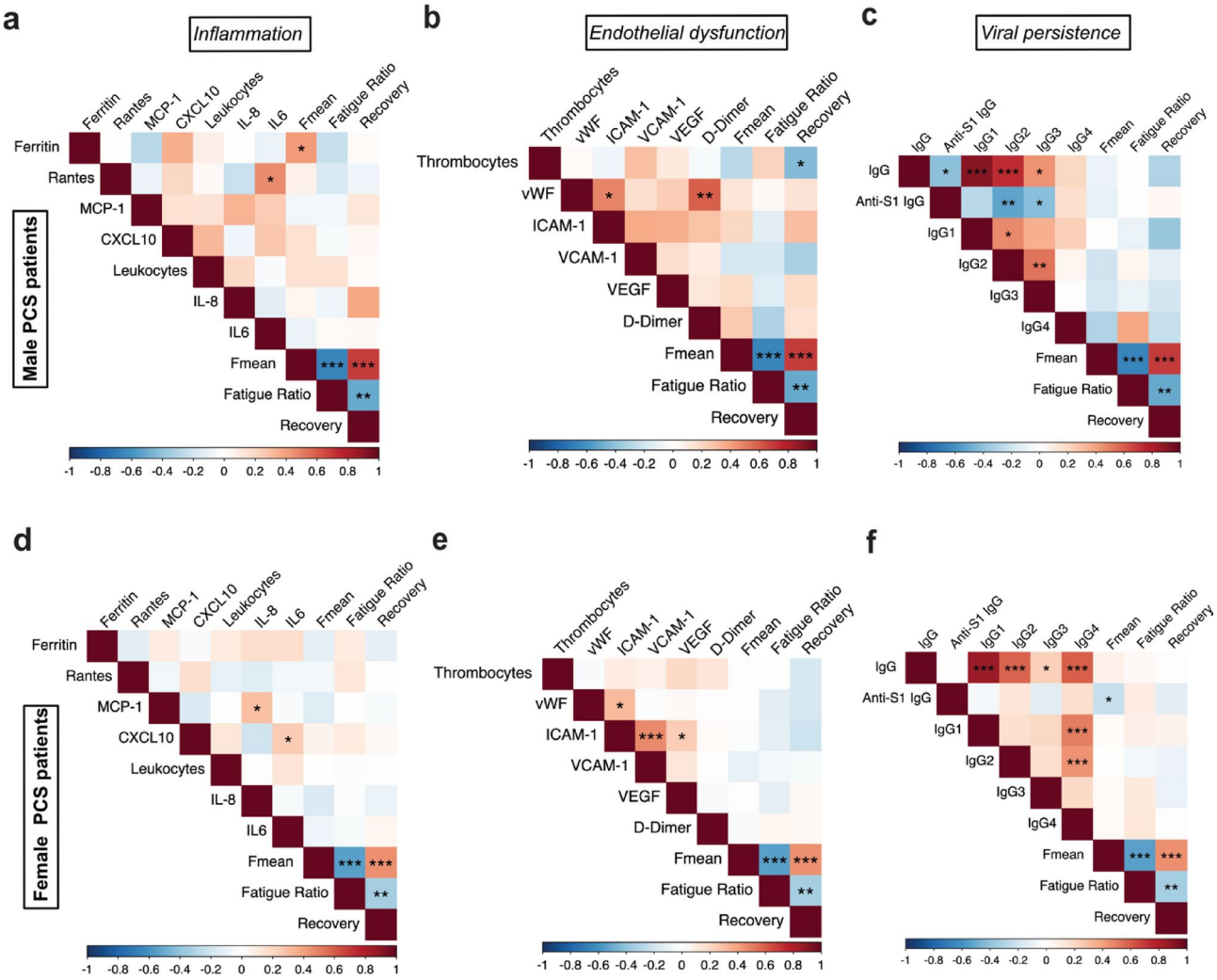


Fig. 3. This correlation matrix plot visualizes the Spearman correlation coefficients for male and female PCS patients between HGS parameters and markers of (**a** and **d**) chronic inflammation (leukocytes (G/l, n = 98), IL8 (fg/ml, n = 101), IL6 (pg/ml, n = 92), MCP-1 (pg/ml, n = 100), Rantes (pg/ml, n = 100), CXCL10 (pg/ml, n = 100)), (**b** and **e**) endothelial dysfunction (thrombocytes (G/l, n = 98), D-Dimer (µg/ml, n = 89), vWF-Ag (%, n = 102), ICAM-1 (pg/ml, n = 86), VCAM-1 (pg/ml, n = 67), VEGF (pg/ml, n = 102)) and (**c** and **f**) viral persistence (Anti-S1 IgG (ng/ml, n = 102), IgG, IgG1, IgG2, IgG3, IgG4 (mg/dl, all n = 99)) for male (**a**–**c**) and female (**d**–**f**) PCS patients. The varying sample sizes for the individual correlations are due to differences in the availability of the respective laboratory parameters, as not all laboratory values were available for every participant. Each cell in the matrix represents the Spearman correlation coefficient between two variables with the color intensity indicating the strength and direction of the correlation. Positive correlations are shown in shades of red, while negative correlations are shown in shades of blue. *P* values were calculated to determine statistical significance of the observed correlations and significant correlations are indicated by asterisks: $p < 0.05$ *, $p < 0.01$ ** and $p < 0.001$ ***. Abbreviations: Anti-S1 IgG = immunoglobulin G against SARS-CoV-2 spike subunit 1; CXCL10 = C-X-C motif chemokine ligand 10; Fmean = mean handgrip strength; ICAM-1 = intercellular adhesion molecule 1; IgG: immunoglobulin G; IL = interleukin; MCP-1 = monocyte chemoattractant protein 1; PCS = post-COVID syndrome; RANTES = regulated on activation, normal T cell expressed and secreted; VCAM-1 = vascular cell adhesion molecule 1; VEGF = vascular endothelial growth factor; vWF = von Willebrand factor.

associated with worse functional outcomes and a more severe phenotype of PCS, corroborating the link between impaired muscle strength and reduced pulmonary function parameters[38].

Although HGS parameters can identify PROMs in PCS patients with and without ME/CFS, we also showed that all measurements of HGS were significantly reduced in PCS patients with a phenotype of ME/CFS. Therefore, we argue that reduced handgrip strength, muscle fatigue, and poorer recovery of muscle strength may represent components of the ME/CFS phenotype, as previously described[10,39]. However, other clusters of PCS may also present impaired HGS, and dynapenia should not be considered a specific marker of ME/CFS. This interpretation is supported by our ROC analyses, which revealed only weak discrimination between PCS

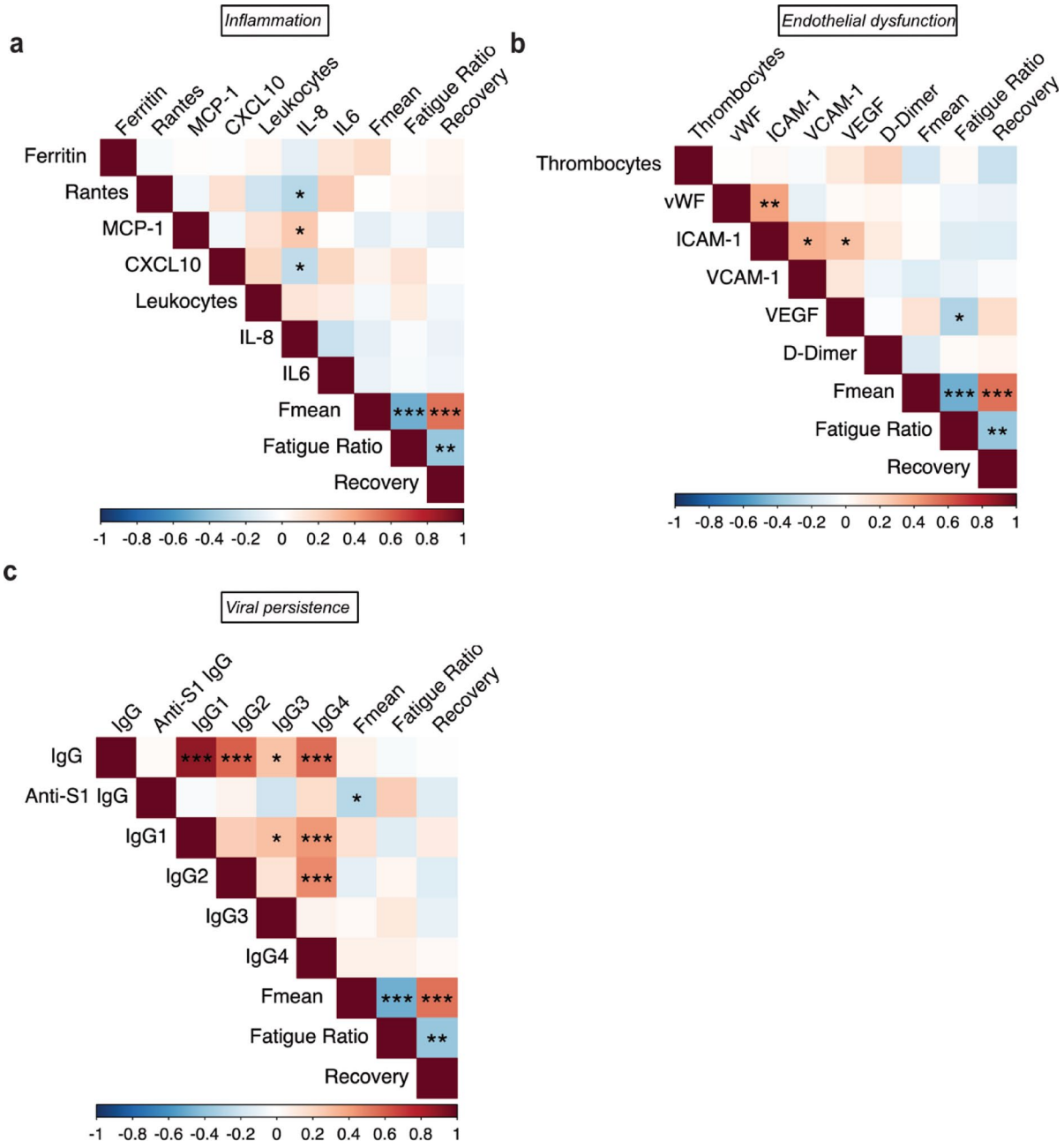


Fig. 4. This correlation matrix plot visualizes the Spearman correlation coefficients for PCS patients with ME/CFS between Fmean (kg), the fatigue ratio and recovery, and markers of chronic inflammation (**a**), endothelial dysfunction (**b**), and viral persistence (**c**). Each cell in the matrix represents the Spearman correlation coefficient between two variables, with the color intensity indicating the strength and direction of the correlation. Positive correlations are shown in red, whereas negative ones are in blue. P values were calculated to determine the statistical significance of the observed correlations, and significant correlations are indicated by asterisks: $p < 0.05$ *, $p < 0.01$ **, and $p < 0.001$ ***. Abbreviations: Anti-S1 IgG = immunoglobulin G against SARS-CoV-2 spike subunit 1; CXCL10 = C-X-C motif chemokine ligand 10; Fmean = mean handgrip strength; ICAM-1 = intercellular adhesion molecule 1; IgG: immunoglobulin G; IL = interleukin; MCP-1 = monocyte chemoattractant protein 1; PCS = post-COVID syndrome; RANTES = regulated on activation, normal T cell expressed and secreted; VCAM-1 = vascular cell adhesion molecule 1; VEGF = vascular endothelial growth factor; vWF = von Willebrand factor.

patients with and without ME/CFS by HGS parameters (Fmean: AUC 0.63, 95% CI 0.52 to 0.74; fatigue ratio: AUC 0.62, 95% CI 0.51 to 0.73; recovery: AUC 0.61, 95% CI 0.50 to 0.73; supplementary Fig. 5).

Whether patients with PCS truly have reduced handgrip strength or a lack of drive leading to lower values cannot be conclusively answered by this study. We examined the relationship between HGS and PCS severity in patients with different degrees of depression severity (as measured by the PHQ-9). PCS patients with more

severe depression had significantly lower Fmean and higher fatigue ratios, which remained significant after controlling for age and gender (supplementary Fig. 6). However, more severe illness is often associated with a greater occurrence of depressive symptoms. Accordingly, it is not possible to judge whether the disease itself or depressive symptoms influence HGS. However, a recorded previous history of mental issues did not affect Fmean in our cohort (supplementary Table 7), and investigations by Appelman et al. demonstrated histologically confirmed muscle damage in PCS patients after induced PEM[40]. Therefore, reduced muscle strength appears plausible as a contributing factor in patients with ME/CFS. To minimize potential motivational bias, participants were familiarized with the procedure, measurements were performed under standardized conditions, and consistent verbal encouragement was provided. These steps reduce the likelihood that observed HGS differences reflect lack of effort rather than physiological impairment. Furthermore, the formation of quotients in the fatigue ratio and recovery could account for motivational effects. These considerations are further supported by the observed associations between HGS parameters and objective biomarkers of endothelial activation and inflammation, as detailed in the following section.

Statistical analysis revealed a significant positive association between Fmean and ferritin levels in male patients (Fig. 3). Interestingly, Kedor et al. reported a similar association, although without adjustment for gender[10]. We hypothesize that this relationship may be linked to iron metabolism, as there is evidence that oxidative stress during muscle activity can impact ferritin levels, potentially as a mechanism to mitigate free radical formation[41]. This could indicate that individuals with less severe PCS and greater muscle strength might retain higher ferritin levels because of lower levels of oxidative stress.

Table 3 shows elevated ICAM-1 levels in patients with abnormal recovery, indicating a potential link between endothelial activation and impaired muscle recovery. We also observed an inverse correlation between thrombocyte levels and muscle recovery in male patients, which could represent a possible trend but requires confirmation in larger cohorts. Increased thrombocytes could signal slight coagulation activation, potentially impacting muscle oxygenation and slowing recovery[42]. Figure 4b shows an inverse correlation between VEGF levels and the fatigue ratio in patients with ME/CFS, which did not remain significant after correction and should therefore be regarded as a possible trend. The role of VEGF in PCS pathogenesis remains controversial, as both elevated and reduced VEGF levels have been observed in convalescent patients[43]. Evidence suggests that dysregulation of VEGF-related signaling pathways in the acute phase of infection may predict the development of PCS[44]. However, in the long term, reduced VEGF levels may also contribute to impaired neuronal tissue perfusion and neurodegeneration[45]. Taken together, our results suggest that ED could contribute to muscle fatigue and weakness in PCS and ME/CFS. However, other ED markers, such as vWF and VCAM-1, showed no clear association with HGS parameters, indicating that the contribution of ED remains uncertain. We believe that muscle weakness in PCS is multifactorial, involving mechanisms beyond inflammation and ED, not captured by our markers. Such mechanisms might include mitochondrial dysfunction, blood vessel amyloid deposits causing local hypoxia, immune response disturbances, viral persistence, and neurological impairments[40,45–52]. PEM is characterized by exercise-induced reductions in mitochondrial enzyme activity, increased amyloid deposits, severe muscle damage, and T-cell infiltration. These muscle abnormalities worsen after the induction of PEM in PCS patients[40,52].

Furthermore, compensatory mechanisms such as epigenetic variability[53] and environmental and lifestyle factors such as physical activity and nutrition[54] may obscure correlations.

The gender ratio in our cohort reflects the current literature, showing an overrepresentation of women[55–57] and supporting the assumption that being female is an independent risk factor for PCS[58]. As we found differences in our subgroup analysis between male and female PCS patients, we believe that the pathophysiology of impaired muscle function may differ.

Patients with higher anti-S1-IgG titers had significantly lower Fmean values. This correlation raises the hypothesis of viral persistence in muscle tissue. However, given that histological processing did not reveal persistence, alternative explanations are more likely[40]. Therefore, another reservoir of circulating S1 seems more likely. Although we identified associations with markers of ED, direct conclusions regarding the pathomechanisms of dynapenia in PCS cannot be drawn. However, using HGS, insights into the PCS phenotype can be achieved. While the usually costly determination of inflammatory markers provides minimal additional information, HGS may help in identifying patients with dynapenia and distinguish them from those with PCS without muscular impairment. In terms of therapeutic strategies, the individual characterization of PCS symptoms could also play a crucial role. Del Corral et al. reported that respiratory muscle training (40 min/day, 6 days/week for 8 weeks) in PCS patients improved quality of life and increased muscle strength and exercise tolerance[59]. However, these patients were not assessed for the presence of ME/CFS or other phenotypes. According to the findings of Appelman et al., such training could lead to a significant deterioration in muscle function. The findings of our study highlight that HGS measurements should accompany any therapeutic interventions for PCS patients.

Strengths and limitations

Our study has strengths and limitations that we would like to address here. The strengths of our study include its prospective design, which allows a more robust assessment of the temporal relationships between variables. Additionally, our cohort is well characterized, enabling detailed analysis of various factors influencing handgrip strength and related parameters. Furthermore, we controlled for age and gender in HGS, ensuring that these variables do not confound our results.

First, the cohort size of 102 patients, while relatively large compared with other cohort-based cross-sectional studies, still limits the statistical power of the applied methods. Moreover, the study was conducted at a single tertiary care center, which may limit generalizability to other settings. Recruitment occurred via social media (76/105) and from the PCS outpatient department (29/105), which may introduce case-mix effects and could result in a cohort with more severe or persistent symptoms than seen in community-based samples. All findings

should be considered exploratory for hypothesis generation and should be validated in larger cohorts. Second, no a priori sample size calculation was performed for HGS; the sample size was determined by the primary outcome of the parent study. The analysis was therefore not specifically powered for the reported HGS outcomes, particularly in the subgroups. Furthermore, HGS was assessed only in the dominant hand and during two short sessions, which may not reflect global muscle strength or long-term fluctuations in performance. Additionally, during the HGS measurements, a learning effect regarding the handling of the dynamometer was observed among the participants, which led to higher average HGS values in the second measurement series than in the first. Therefore, we decided to use the second measurement series for calculating the fatigue ratio and Fmean. This could also be the reason for the small number of significant results related to recovery. Given that no established threshold for recovery ratio exists, we dichotomized the recovery ratio using a cutoff of 0.8, which approximately corresponds to the lower quartile of the distribution in our cohort. This data-driven approach was chosen to identify patients with the most pronounced impairment in muscle recovery and should be considered exploratory. Sensitivity analyses using alternative cutoffs (0.75 and 0.85) yielded similar patterns of association (supplementary Table 8), suggesting that the findings are robust but require confirmation in independent cohorts.

Conclusion

HGS offers a clinical parameter for a better understanding of the PCS phenotype. Patients with pathological HGS measurements should receive special consideration in rehabilitation, as muscle training could pose a risk for these individuals and rehabilitation strategies may need to be adapted accordingly. However, larger cohorts are necessary to determine appropriate thresholds for HGS parameters. Additionally, HGS parameters should be used to monitor therapeutic success. Based on our findings, we cannot draw definitive conclusions about the pathophysiology of dynapenia in PCS, although inflammation markers appear to play a relatively minor role, whereas markers of ED correlate with various HGS parameters. Further studies could correlate histological features with HGS parameters to demonstrate dynapenia at the histological level. Moreover, investigations into endothelial and mitochondrial dysfunction could provide valuable insights into the pathophysiology of dynapenia in the PCS.

Data availability

The data underlying this manuscript will be shared upon reasonable request to the corresponding author. All shared data will be fully anonymized/de-identified to ensure participant privacy in accordance with institutional and GDPR regulations.

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Author contributions

This study was conceptualized and designed by T. W. and C. S. The statistical analysis was conducted by T.W., M.B.-W. and E.T., with valuable support from B.H. Data collection and patient surveys were carried out by A.W., J. N., and V. K. The manuscript was drafted by M.B.-W. All authors reviewed the manuscript. T.W. and M.B.-W. contributed equally to this work.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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